# Coastal Shoreline Change Toolkit

An interactive, form-based demonstration of coastal shoreline change using Copernicus Sentinel-2 surface-reflectance imagery accessed through Google Earth Engine (GEE). The application starts with the supplied Cape Cod study area and lets you compare annual observations, inspect false-colour imagery, and explore waterline movement along cross-shore transects.

# Coastal change and why study it?

Coastlines are dynamic boundaries shaped by waves, tides, storms, currents, sediment supply and human activity. Beaches and dunes can retreat or build out between seasons and during individual storms. Measuring those changes helps coastal communities and researchers understand shoreline trends, identify vulnerable places, assess how beaches and barriers respond to storms, and make better-informed decisions about habitat, infrastructure and coastal planning. No single satellite observation tells the whole story: repeat observations and local context are essential.

Cape Cod is a useful demonstration site because its exposed Atlantic shoreline, long sandy beaches, dunes and barrier features make it a compelling setting for observing waterline variability. Its varied coastal forms also make a useful reminder that one index threshold or transect orientation will not work equally well everywhere.

# What this toolkit can show

The toolkit turns selected Sentinel-2 years into annual median image composites, SWIR/NIR/green false-colour views and annual shoreline outlines. Cloud, cloud-shadow, cirrus, snow, no-data and defective pixels are masked before the median is calculated; the reducer ignores masked values, and locations with no clear observation that year remain masked. No values are filled from other years. The study-area map in Section 2 uses OpenStreetMap for orientation; the imagery results map in Section 4 omits the basemap so the shoreline outlines are easier to inspect, while the separate transect plan map includes OpenStreetMap. Results maps show a different-coloured, unfilled vector shoreline outline for each observation year; the colour key sits beside the imagery map. A separate plan map shows the cross-shore transects used to compare waterline positions in metres along each transect on a linear scale.

Together, these views support an accessible first look at where the visible waterline has shifted, how that signal varies along the study area, and how results compare between selected years. They are intended for exploration and communicationâ€”not as a substitute for tide-corrected shoreline mapping, field surveys or a validated sediment budget.

> **Interpret results with care.** Optical imagery detects a waterline proxy, not a datum-corrected shoreline. Tides, waves, season, clouds and the selected water-index threshold can all affect the result. The volume figure is an assumption-based screening estimate, not a measured sediment budget.

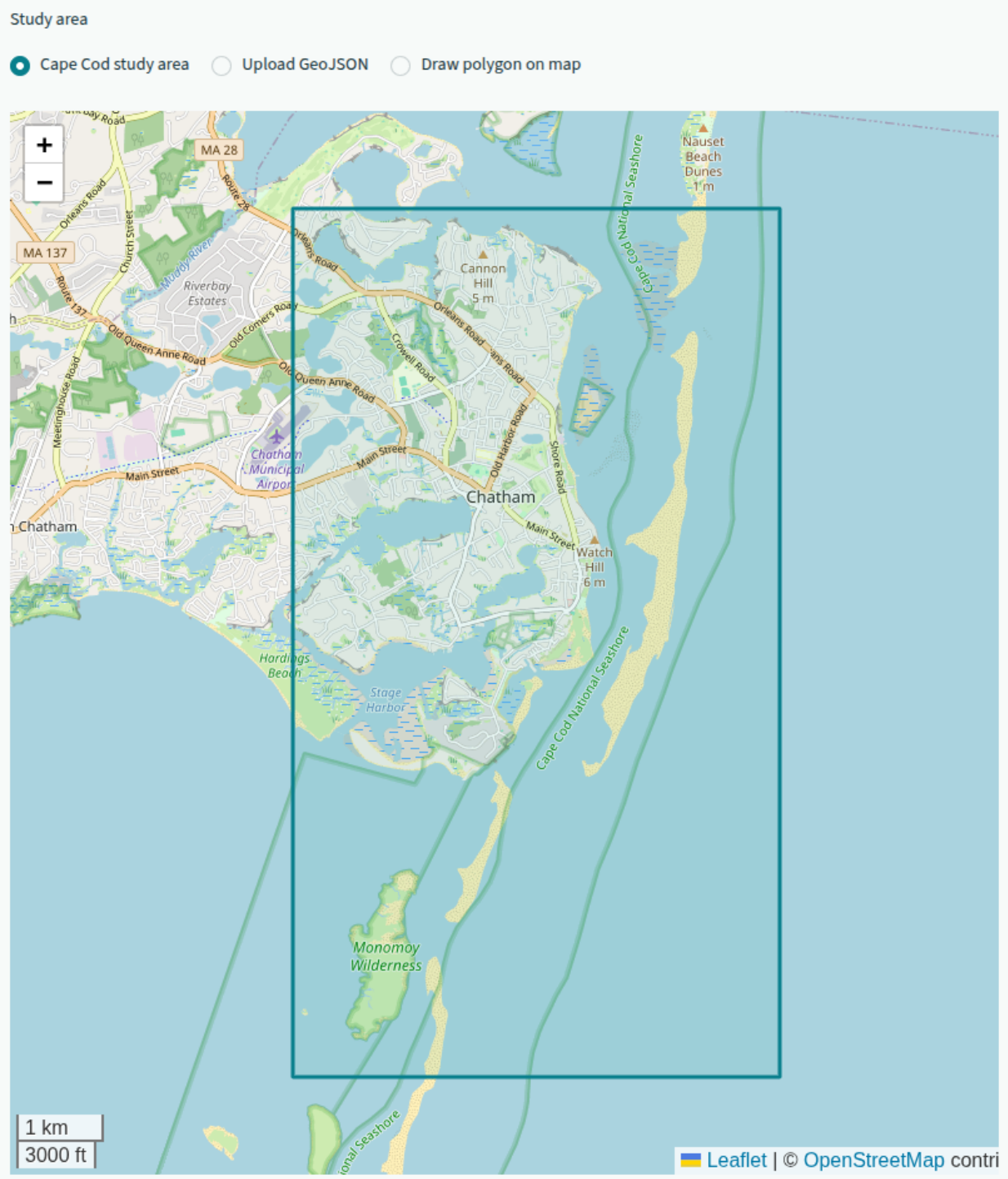
# What the app does

- Guides users through Earth Engine setup before they start an analysis.
- Includes the Cape Cod GeoJSON study area; also accepts uploaded GeoJSON or a polygon drawn on the map.
- Builds annual, cloud-masked Sentinel-2 median composites at a selectable interval, every two years by default.
- Shows SWIR/NIR/green false-colour imagery and year-coloured, unfilled vector shoreline outlines.
- Places the annual shoreline-colour key beside the imagery map.
- Uses the supplied 12 transects for the default Cape Cod study; custom study areas use generated transects. A separate plan-view map shows the lines used.
- Compares waterline crossings along projected, evenly spaced transects and provides CSV and GeoJSON downloads.
- Optionally samples a local, projected GeoTIFF DEM to display static elevation profiles along the same transects.
- Calculates a signed planform-change proxy and a clearly qualified volume estimate using a user-supplied active-profile height.

# Example outputs

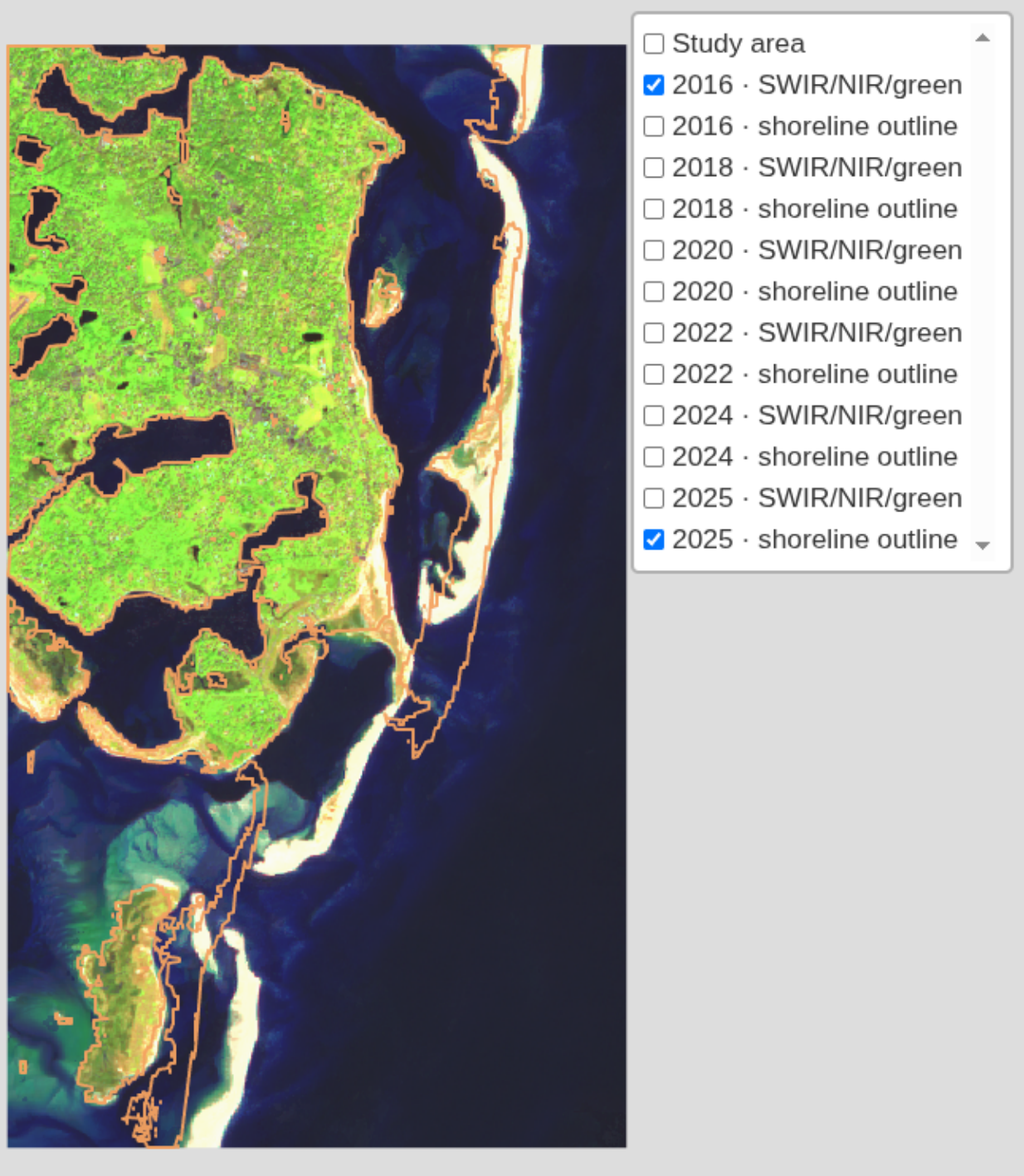
The screenshots below show example views from the app using the Cape Cod study area. They illustrate the study-area selection, Sentinel-2 imagery and annual shoreline vectors, the supplied transects, and the transect-based summaries.

## Select and inspect the study area

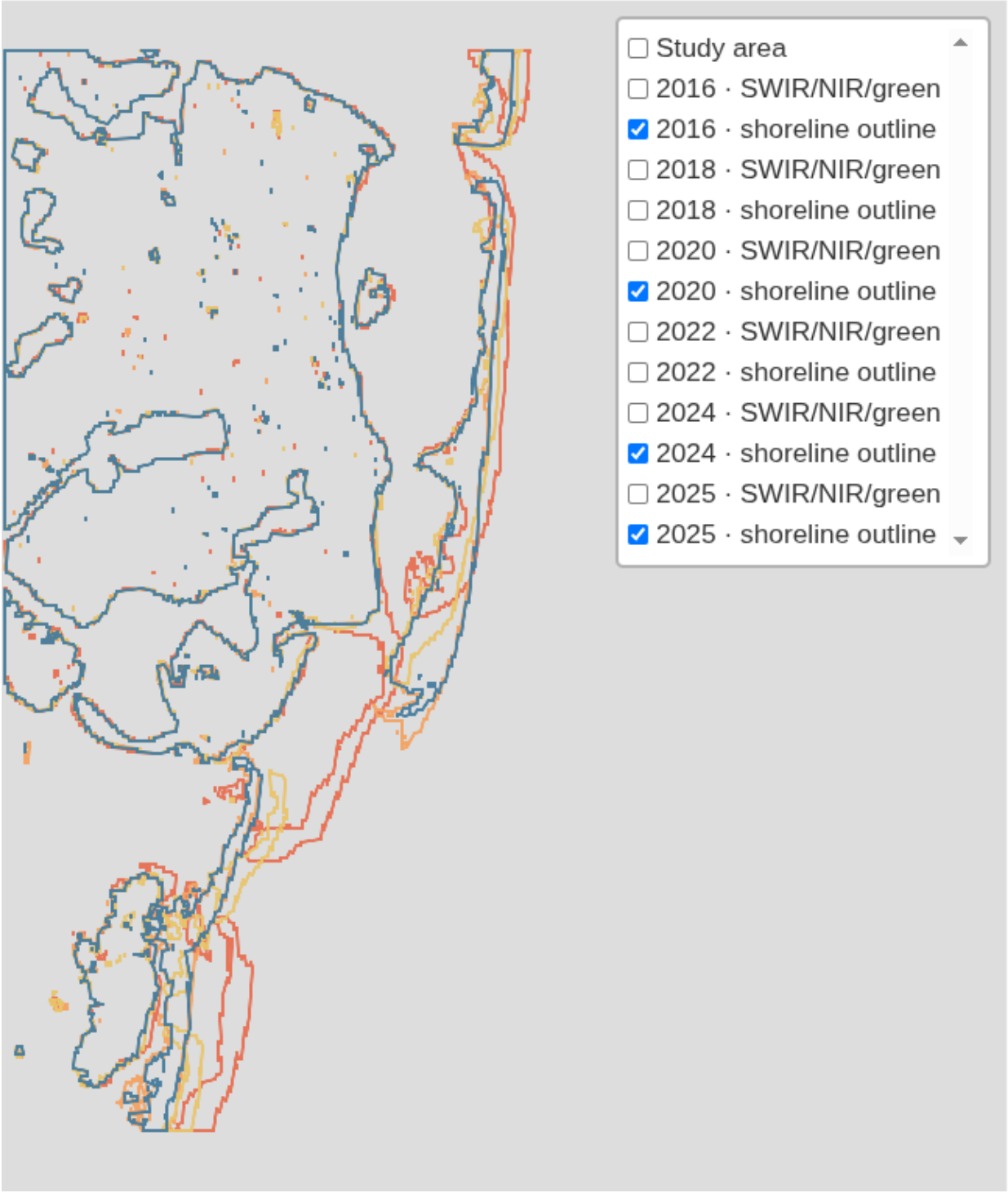



The polygon defines the area processed by Earth Engine. Users can keep this default area, upload another GeoJSON polygon, or draw an area on the map.

## Compare imagery and shoreline outlines




The false-colour composite provides image context for a year-specific, unfilled shoreline vector. Use the map layer control to switch observation years and compare outlines.

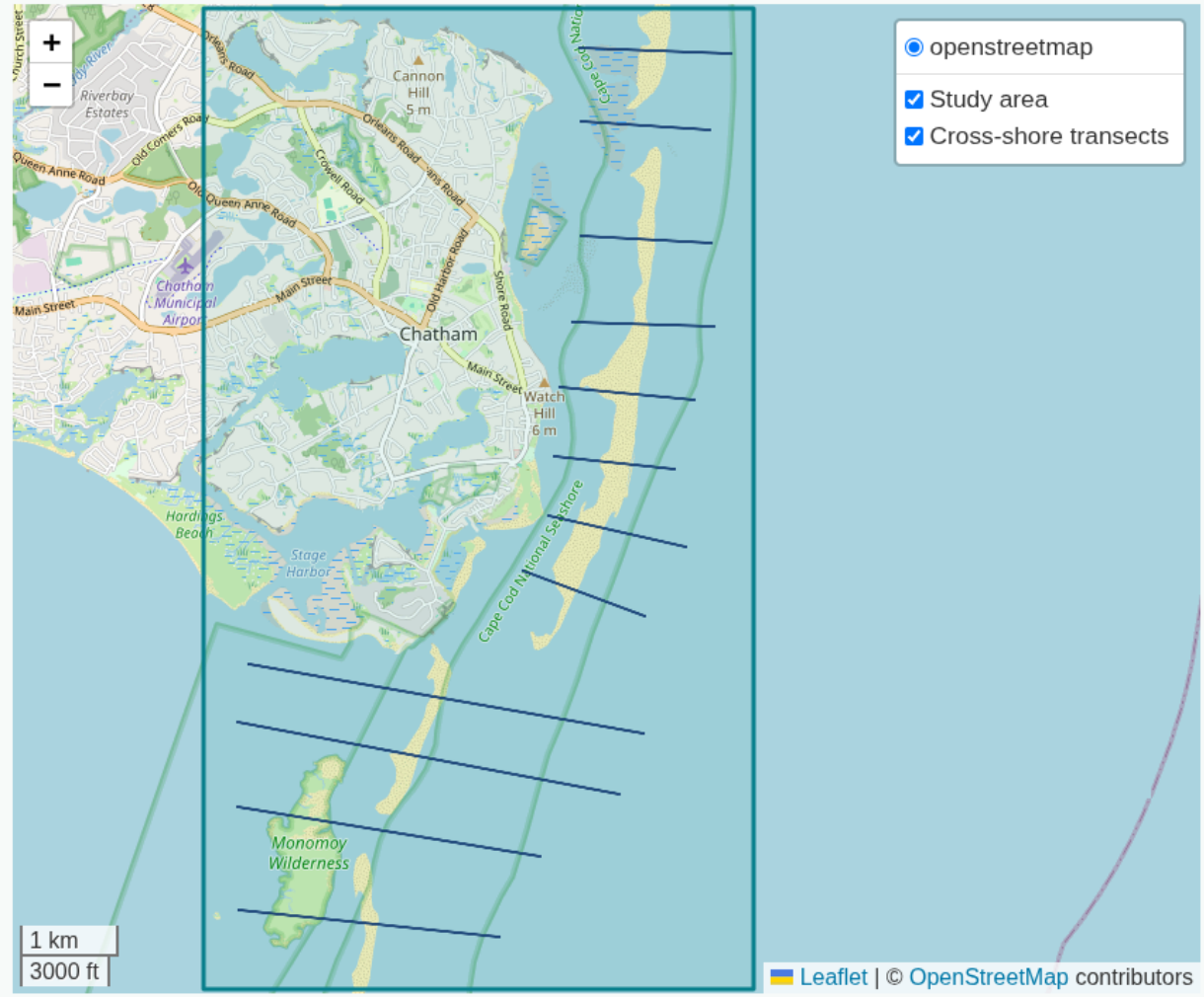



The overlay shows annual shoreline outlines in distinct colours so their positions can be compared without opaque land-mask fills.

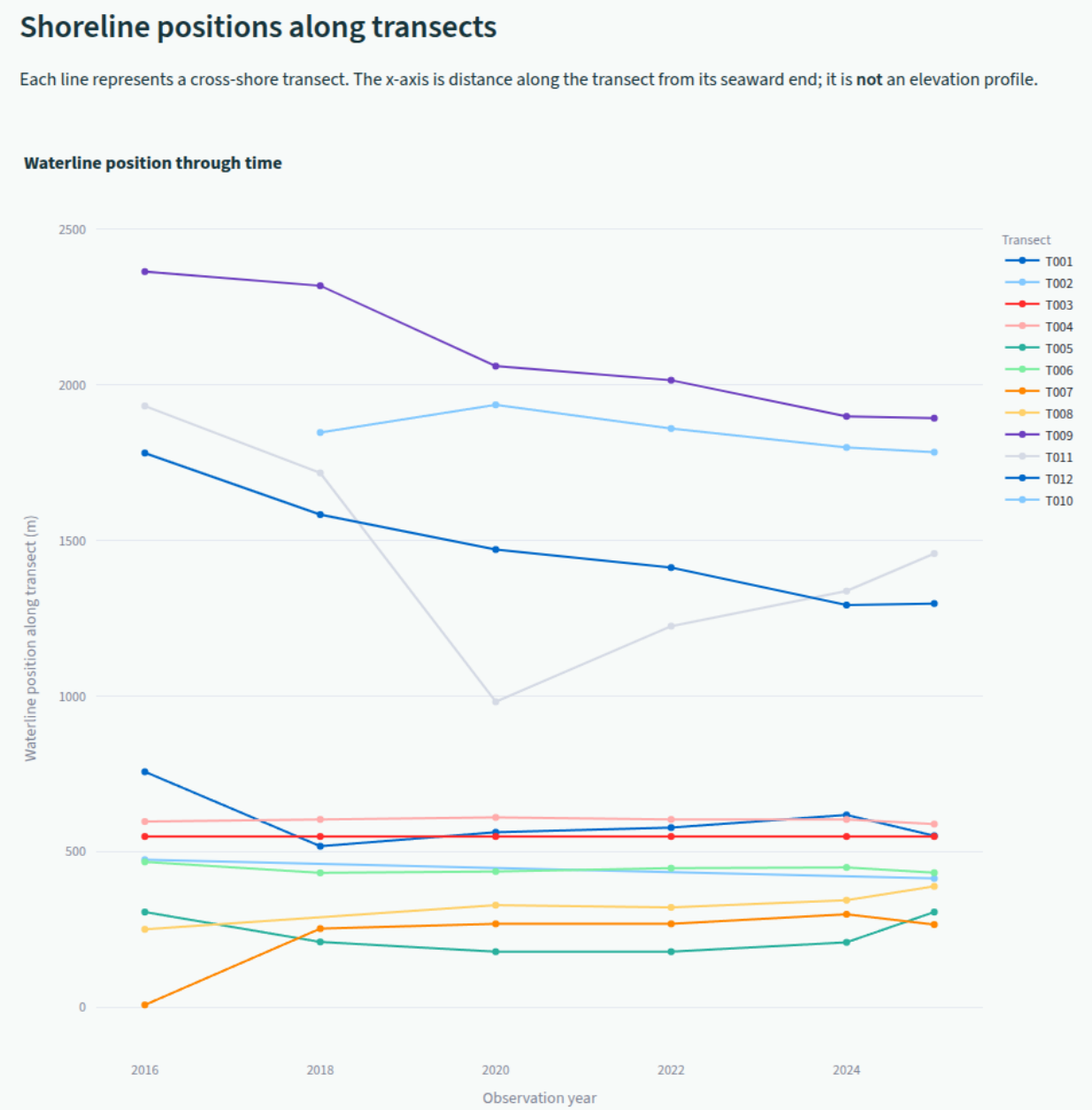
## Inspect transects and waterline positions

## Transects in plan

Supplied cross-shore transects used to compare waterline positions. Shown separately from the imagery map.




The default Cape Cod analysis uses the supplied cross-shore transects. The separate plan view makes their orientation and coverage visible.




Each line tracks the detected waterline position along one transect across the selected observation years. This is a linear distance measure, not an elevation profile.

## Review the indicative volume summary

| Transects compared | Signed planform-change proxy | Indicative volume change |
|---|---|---|
| 11 | -1,377,047 $m^2$ | 6,885,235 $m^3$ |

The displayed volume is a screening estimate based on planform change and the user-supplied active-profile height; it is not a measured sediment volume.

# Requirements

- Python 3.10 or newer (the supplied `.venv` uses Python 3.12).
- A Google account with access to Google Earth Engine.
- A Google Cloud project with the Earth Engine API enabled.

# Set up a clone

From the project directory, create and activate a virtual environment, then install the project dependencies:

```
bash python -m venv .venv
```

Linux or macOS:

```
bash source .venv/bin/activate
```

Windows PowerShell:

```
powershell .venv\Scripts\Activate.ps1
```

Install the requirements and the local package:

```
bash python -m pip install --upgrade pip python -m pip install -r requirements.txt python -m pip install -e .
```

# Connect to Earth Engine

1. Enable the Earth Engine API for a Google Cloud project and note its project ID.
2. With the virtual environment active, authenticate in the same terminal:

```
bash earthengine authenticate --auth_mode=localhost
```

1. Start the app:

```
bash streamlit run app.py
```

1. Enter the Cloud project ID in the app and select **Verify Earth Engine connection**.

The app uses the Earth Engine credentials configured for the current user. It does not store service-account keys or credentials. Optionally set `EE_PROJECT` before launching Streamlit to prefill the project ID.

# Run an analysis

1. Authenticate and verify the Earth Engine connection.
2. Keep the included Cape Cod area, upload a GeoJSON Polygon/MultiPolygon, or select **Draw polygon on map** and draw the study boundary. GeoJSON coordinates must be longitude/latitude (EPSG:4326); the maximum area is 500 kmÂ¸.
3. Select observation years from 2016 onward and the interval. The default interval is two years. The app builds one annual median composite per selected observation year after masking cloudy pixels.
4. The default Cape Cod study uses the supplied cross-shore transects in `data/transects/transects.geojson`. For uploaded or drawn study areas, set transect spacing and the bearing **from sea toward land**.
5. Optionally upload a single-band, projected GeoTIFF to display static elevation transect profiles.
6. Select **Run shoreline comparison**. Toggle false-colour imagery and year-specific shoreline-outline layers using the map's layer control, then inspect the separate transect plan map and download the transect results.

The default AOI is in `data/aoi/cape cod.geojson`. For details on the index, shoreline proxy, assumptions and limitations, see `docs/methodology.md`.

# Project structure

text . â”śâ”€â”€ app.py # Streamlit interface â”śâ”€â”€ data/ â”‚ â”śâ”€â”€ aoi/ â”‚ â”‚ â””â”€â”€ cape_cod.geojson # Supplied Cape Cod polygon â”‚ â””â”€â”€ transects/ â”‚ â””â”€â”€ transects.geojson # Supplied Cape Cod transects â”śâ”€â”€ docs/ â”‚ â””â”€â”€ methodology.md # Processing choices and interpretation â”śâ”€â”€ src/ â”‚ â””â”€â”€ coastwatch/ â”‚ â”śâ”€â”€ analysis.py # Transects and change estimates â”‚ â”śâ”€â”€ config.py # Shared settings â”‚ â”śâ”€â”€ gee.py # Earth Engine imagery and land mask â”‚ â”śâ”€â”€ geometry.py # AOI validation and transect geometry â”‚ â””â”€â”€ raster.py # Optional local DEM sampling â”śâ”€â”€ tests/ # Local unit tests â”śâ”€â”€ pyproject.toml â””â”€â”€ requirements.txt

# Development checks

```
bash pytest
```

# Data and attribution

Sentinel-2 surface-reflectance scenes are accessed from the `COPERNICUS/S2_SR_HARMONIZED` Earth Engine collection. The app does not bundle or redistribute satellite imagery. Respect the applicable Copernicus data terms and Google Earth Engine terms when using or sharing derived outputs.

# License

See LICENCE.